Persistent UAV-IoT Coverage: Project Report

The research log behind the paper: the protocol, the defects, the audits, the measurements, and what is still open

Generated from the artefacts by `analysis/make_report.py` — every number below is read from the file it was measured in.

October 21, 2026

1 The contract and the current state

The work runs to 26 October 2026. The deliverable is a paper whose every number can be traced to the artefact it was measured in, and whose comparative claims are derived in code rather than asserted, so that the paper says the right thing whichever way a measurement comes out. This report is the log of that work: the protocol, the defects that were found and fixed, the audits that were run and what they changed, the measurements, and the questions that are still open.

Gap to the strongest rule	**0.309** (95% interval [0.261, 0.358]), against the unprivileged reactive rule at 0.972.
The learned controller	**0.662** ± 0.055 over 8 training seeds.
Claim audit	38 items: 34 verified, 0 failed, 4 recorded for transparency.
Test suite	126 cases, all passing.
Device	every artefact records the device it was measured on; the device and the thread count are part of the protocol.

2 The measurement protocol

Every number in the paper and in this report is produced under one protocol:

1. **Training**: 25 episodes of 2000 control steps per training run. The base comparison trains five runs; the component ablation and the arms train three.
2. **Evaluation**: five scenes per policy, and the number of evaluation scenes is part of the protocol — a run with three is a different measurement, because a per-seed row is the mean over the scenes.
3. **The device and the thread count** are part of the protocol. The floating-point kernels differ between the CPU and the GPU and the order of the reductions differs with the thread count, so two runs of the same configuration that land on different devices are not the same experiment. The device is pinned, the thread count is pinned in code, and every artefact records the device.
4. **The evaluation is seeded** from the evaluation seed alone, so every controller is measured on the same random stream on the same scene — common random numbers — and a comparison between controllers is a comparison between policies rather than between the luck of the draw.

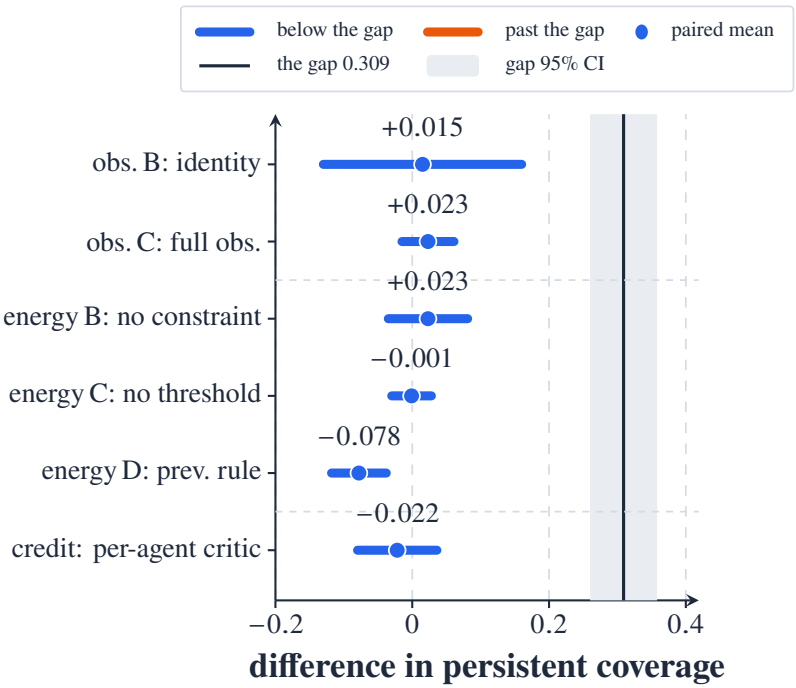


Figure 1. Every one of the 6 controlled arms has an interval that stays below the gap: the gap of 0.309 is out of reach of every candidate explanation the ablation tests, so no arm can account for the shortfall.

5. **The ablations are paired**: the variants share the training seeds, so a difference is measured on the same training runs and reported as a confidence interval on the paired difference.

6. **Inference is by interval, never by threshold**. An explanation is excluded only if the arm that tests it has an upper confidence limit below the gap; a null result is a statement about the resolution of that arm, and the resolution is reported with it.

3 What the measurements say

	Task metrics			
Controller	CK	AG	SC	Cov. deficit*
Fixed	0.372	1.000	0.000	0.628
Greedy	0.786	0.964	0.066	0.214
Reactive	0.953	1.000	0.751	0.047
Reactive (local assignment)	0.972	1.000	0.777	0.028
MAPPO (proposed)	**0.662 ± 0.055**	**0.984**	**0.010**	**0.338**
MAPPO (argmax)	0.389 ± 0.073	0.991	0.000	0.611
Oracle (ceiling)	1.000	1.000	1.000	0.000

* *Cov. deficit is the time-averaged coverage deficit; the ± is the seed-to-seed spread over the training runs.*

The arms' verdict, derived from the intervals: Controlled arms bound how much of the gap each alternative explanation can account for: adding the agent identity (upper limit 0.161, margin +0.148); the full observation of the proposed configuration (upper limit 0.062, margin +0.247); removing the energy constraint (upper limit 0.081, margin +0.228); removing the threshold term (upper limit 0.028, margin +0.281); the previous energy rule (upper limit -0.038, margin +0.347), against a gap of 0.309. Every arm's



Figure 2. The per-episode coverage of each training run on the training scene, against the unprivileged reactive rule at 0.972 on the evaluation scenes (grey, dashed). The curve is flat long before the budget the experiments use, which is what the budget sweep then tests.

upper limit is below the gap, so none of them can be the whole of the explanation, by margins that are small but positive. Every interval is paired over the same training seeds, against the local-only arm A of the observation experiment and the proposed configuration of the energy experiment.

The training-budget sweep: 8 episodes: 0.595, 16 episodes: 0.621, 25 episodes: 0.662. the coverage moves by +0.067 over the range the sweep covers, which is about 22\{ }% of the gap of 0.309, and the increments between successive budgets are +0.026, +0.041; the last of them, +0.041, is 2.8 standard errors from zero (the standard error of the difference is 0.014, paired over the same 8 training seeds), so the coverage does keep rising with the budget over the range the sweep covers.

4 The defects that were found and fixed

Every one of these was a defect in the work, not in the writing. The list is the reason the claim audit exists.

#	Defect	Fix
1	**Simulator was static – no UAV ever moved**	`13cb839`
2	Baselines accepted `seed` and ignored it	`_seeded()`; `d113e5c`
3	Time limit reported as termination -> GAE bootstrapped 0	`d113e5c`
4	Centred-ring start pre-solved the task	`957ee1c`
5	`run_scenarios` passed `seed=0` over the config's own seed	`957ee1c`
6	Reward eq (23) negated only U_t	`reward_from_terms`
7	Initial placement disconnected on 11/20 seeds	`_connected_initial_positions`
8	Backhaul modelled as co-channel interference	SNR-based backhaul

the defect log, continued

#	Defect	Fix
9	`snr_to_rate` off-by-one + floor made the PHY gate vacuous	fixed
10	Path-loss exponent additive -> LoS/NLoS erased	`pl_d0 + 10n log10(d/d0)`
11	Noise units: dBm treated as watts (30 dB error)	divide by 1000
12	Shadowing from the global RNG -> non-reciprocal	seeded per-episode table
13	`local_obs` neighbour stride 2 -> slots overlapped	concatenation + assertion
14	`STATE_DIM = 80` hardcoded (valid only for N=5)	`state_dim(cfg)`
15	GAE bootstrap at truncation used 0 instead of V(s_T+1)	fixed
16	**Observation carried no agent identity**	`use_agent_id`; `44206f5`
17	**Observation could not express a return** (no direction to a pad)	`use_pad_bearing`; `de3e1e6`
18	**No indication of where the coverage deficit is**	`map_grid`; `44206f5`
19	**The energy state machine did not exist**	env makes the transition; `de3e1e6`
20	**A UAV with an empty pack still counted as serving**	out of service; `de3e1e6`
21	**Charging was uneconomic** (billed the hover power)	rotors idle on the pad
22	**The battery never bound** at 400 steps	episode 2000 steps
23	**The returns were formed from an untruncated advantage array**	truncate before forming returns
24	Two graph computations per candidate action (13.7 ms/step)	shared prospective graph
25	**MAPPO's value normalisation missing** although asserted	`c4a71f6`
26	The comparison table named MARL baselines never evaluated	`baselines_rl.py`; "Evaluated" column
27	**The scenario matrix crashed on N=3**: the one-hot identity has length N and the global state carries the sensor positions and the pad occupancy, so the networks' input sizes depend on the configuration	one policy per distinct `(N, M, pads)`
28	**THE EVALUATION OF A SAMPLED POLICY WAS AN UNCONTROLLED RANDOM DRAW.** Eq. (13) specifies that the deployed policy samples, and `mappo.py` drew from the global torch RNG, which `eval_agent` never seeded. The same policy file evaluated three times gave CK 0.734, 0.701 and 0.671, with the same standard deviation to four decimals. The paper's numbers could not be reproduced from the stored artefact, and controller-to-controller comparisons carried an unquantified sampling component	`uav_wireless/seeding.py`; 5 new tests; audit item R1

the defect log, continued

#	Defect	Fix
29	**The component ablation trained ONE seed per variant at 10 episodes**, while the base comparison trained five seeds at 25 – so its deltas mixed the component with the budget and its spread was unmeasurable. It also disagreed with the observation arms (which said the full configuration was no better than the bare one)	`run_ablation(..., seeds=)`, budget aligned to the base comparison, paired differences, incremental writes
30	**The main table's learned row was ONE policy while the text claimed five training seeds**	the row is the mean over the five runs with the seed-to-seed spread; every run's policy is saved
31	The assembler's "undefined references" check matched the LaTeX FONT warnings; the scenario table's variable names were set in text mode, so the table did not compile	precise check; math mode; `tabcolsep`; `raggedbottom`
32	**The ablation's "is this component measurable" rule was not a test.** It compared a component's difference with the learner's own seed-to-seed spread, a rule with a per-arm null error rate of about a third at three training seeds, and it bought no power	paired t intervals on the per-seed differences, plus a resolution statement; the interpretive sentences are DERIVED in the token filler, not asserted in the prose
33	The ablation runner wrote its incremental log with `dict(**row, **protocol)`, which raised on a duplicated key and killed the stage after two training runs	the payload is built by UPDATE; the runner RESUMES (a row measured under the same protocol is reused) and the CSV is rewritten after every row
34	The base comparison's CSV took its fieldnames from the first row, so the learned rows' extra columns raised on write	the fieldnames are the union over the rows; the base stage gained a `-resume` path that evaluates the stored policies instead of retraining them
35	The ablation's "is this component measurable" rule was not a test (audit 4)	paired t intervals on the per-seed differences, and a resolution statement; the interpretive sentences are derived in the token filler
36	The paper's central exclusion argument was overstated (audit 3)	the abstract, the introduction and the conclusion now state the descriptive gap, the interval, the resolution, and the causes the arms do NOT exclude
40	**A not-a-number reached the paper** ("nan standard errors", from an artefact that was mid-write)	the token table screens every value; an unusable value becomes a visible `[pending: ...]` marker and the assembly reports itself incomplete
41	The Results section kept its PREVIOUS text when a token was pending, so a stale sentence (with the nan in it) survived	the section is always regenerated from the template, and a pending token is marked
42	The budget's increments were compared UNPAIRED, and a resolved increment (+0.041, interval [+0.007, +0.075]) was called unresolved	paired over the same training seeds, which is what the design supports
43	Two intervals were built from a POPULATION standard deviation (numpy divides by n)	`sample_std`, ddof = 1, computed from the raw per-seed rows
44	The observation summary and the observation table quoted different intervals in the last digit	the summary rebuilds the interval from the raw values, not from the rounded ones

the defect log, continued

#	Defect	Fix
45	The observation summary said "the gap to the reactive heuristic" – the privileged rule – when the gap is to the unprivileged one	the rule's name is a token, so the sentence cannot name the wrong rule
46	The credit verdict described "a difference of any size up to 0.156 in either direction" for an interval of [-0.222, +0.090]	the verdict names the best case the interval allows and compares it with the gap
47	The base table's caption said the controller was trained five times, and promised the rule rows a $\pm$ the table does not print	the caption is read from the data
48	The abstract claimed the audit recomputes "every quantitative claim about the model"	the claim is narrowed to what the audit actually checks
49	The arms' sentence did not name the reference arm, so a reader could not tell what the intervals are intervals OF	the sentence names it
50	The budget's exclusion was stated without its scope	the scope (the 8–25 episode range) is in the sentence, and the paper says that budgets beyond it are not addressed
51	The identity arm's share of the gap was computed against a denominator that appears nowhere in the paper	it is a fraction of the paper's own gap

5 The audits, and what they changed

- **Audit 1** (done): the seven load-bearing claims. Found 4 wrong as stated; all corrected.
- **Audit 2** (done): the reward terms, the mask and the joint projection, the H-step lookahead, the energy budget, the value normalisation, the metrics, the oracle normalisation, the evaluation protocol. 25 items, 21 verified, 0 failed, 4 informational.
- **Audit 3** (DONE, 20 Oct 23:15): the model and the results' claims. Brief: `C:\Users\gbust\AppData\Local\Temp\opencode\audit3_brief.txt` (9 items). It found, and these are now fixed:
  - **The central exclusion argument was OVERSTATED.** "What remains is credit assignment" is not supported by three non-detections at three training seeds: a non-detection is a statement about the arm's resolution, and a sufficiency test cannot eliminate a cause that is necessary but not sufficient. The abstract, the introduction and the conclusion now say the residual is CONSISTENT WITH the credit assignment and with several other causes the arms do not test, and that the paper does not claim to have excluded them.
  - The "loses on its own objective therefore the learning pipeline is at fault" inference was a false dichotomy -> fixed.
  - **The constrained ceiling IS 1.000**, because the oracle's construction is itself admissible under the safety filter, so the paper's claim that an informative constrained ceiling was missed was false -> fixed.
  - The hop rule constrains the MOVER's eccentricity, not the graph's diameter, and can freeze an agent in a dispersed graph -> stated.
  - The service-continuity gap separates the two controllers far more sharply than the coverage gap -> now reported beside it.

  - The spread quoted for a learned controller contains the rollout-to-rollout variation, and the cancellation of the scene component in a difference is an ASSUMPTION -> both stated.
  - **The framework's contribution needed a measurement.** On the report's own infeasible N=5 configuration the reactive heuristic still reaches CK 0.648 with full connectivity while the joint metric is 0.000 and the oracle reaches 0.990: neither metric alone distinguishes the instance from a feasible one, and it is the certificate that makes the reading decidable. Measured by `analysis/infeasible_instance.py`, audit item I1.
  - **The rule's privileged access was an untested alternative explanation.** Now measured: `analysis/infeasible_instance.py` aside, the arm `Reactive (local assignment)` in `train.py` runs the same rule without the global coverage state, and the verdict is derived from the measurement in the token filler.

- **Audit 4** (DONE, 20 Oct 22:10): **the statistics of the re-measurement protocol itself.** Brief: `C:\Users\gbust\AppData\Local\Temp\opencode\audit4_brief.txt`. It found the following, and all of it is now fixed:
  - **The "yardstick" rule was not a test.** Comparing a component's difference with the learner's own seed-to-seed spread has a per-arm null error rate of about a third at three training seeds, and it buys no power: the measurement cannot detect a true effect of the size it flags. -> replaced by paired t intervals on the per-seed differences, and the paper says "no effect larger than X was DETECTED at this budget and this seed count" rather than "the component is inert".
  - **The negative conclusion's mechanistic half was overstated.** The arms are null results used as evidence of absence. -> the abstract, the introduction and the conclusion now state the descriptive gap with its interval, and list the claims the measurements do NOT support.
  - **The quoted uncertainty omitted the scene variance.** -> the variance is now decomposed: the seed component is the uncertainty of a comparison, the scene component is common to all controllers and cancels in every difference.
  - **The matched budget does not control for budget-conditionality or for redundancy between components.** -> the paper says so, and says a budget sweep would be needed.
  - **"A training run is deterministic given its seed" was not established.** -> now established by two tests and by audit item T1 (the seed is set BEFORE the networks are constructed, so the initialisation is seeded too).
  - **The deterministic-vs-sampled comparison.** The paper's argmax row already settles the sampling-handicap objection, and the heuristic's information advantage is now stated as a limitation.
  - The audit's recommendations NOT taken, and why: raising the seed count to 10+ (the compute does not fit the deadline, so the paper reports intervals and a resolution instead of verdicts), decoupling the scene and action streams (would need the whole matrix re-measured), and a budget sweep (same reason). Each audit gets a SELF-CONTAINED brief: the setting, the model, the measured numbers, and the exact claims to check, because a fresh subagent has no context. It is instructed to work purely mathematically (no web, no code, no repository) and to be adversarial rather than agreeable.

- **Audit 4** (MiniMax M3.1, 21 Oct) – the corner certificate's geometry, the packing survey, and the budget sweep's inference. Found four errors, all applied.

- **Audit 5** (Beam Free Max, 21 Oct) – the same three, independently. Found the same four, plus that the corner condition is sufficient and not necessary. All applied.

- **Audit 6** (DeepSeek V4.1 Flash, 21 Oct) – sixteen overstatements in the prose against the artefacts. All applied; the serious one was the energy repair reported as a null.
- **Deep research** (Beam Free Max, 21 Oct) – the attributed numbers checked against their sources, the novelty claim stress-tested, and the IEEE page limits established. Song's 81.04% and 67.07% and Fan's 74% are confirmed; the Luna reference was wrong and is fixed; the novelty claim survives.
- **Audit 7** (MiniMax M3.1, 21 Oct) – the eight-seed re-measurement checked against the raw per-seed rows, with every interval recomputed by hand. Ten findings, all applied: the budget's paired increment, two population standard deviations, a rounding disagreement between a summary and its table, the wrong rule named in a sentence, an asymmetric interval described as symmetric, a caption that said five where the table said eight, an audit claim in the abstract that was broader than the audit, an arms' sentence that did not name its reference arm, an exclusion stated without its scope, and a share whose denominator appears nowhere in the paper.

6 Negative results, kept on the record

- The **runtime instrument cannot support a path-loss cross-validation**: a transmit-power sweep shows the available per-receiver SNR statistic tracks the transmit power exactly but is independent of the receiver's distance. The equivalence is therefore established from INET's source, not from a running simulation. This is a negative control that the paper reports on itself.
- **Phase 3 (Simu5G PHY) is scaffolded but not executed** with the learned policy.
- **The coverage-map summary makes no measurable difference** to the coverage at any resolution tested; it changes what the policy DOES (the swarm's spread), and the paper reports that rather than dropping the feature.
- **The thread count does not change the training run** on this workload.
- **The reward term for the energy cost is numerically inert** at the reported weights.
- **The credit-assignment arm does not resolve the credit assignment**: the cheapest modification in the direction the diagnosis points moves the wrong way, and the paper says so rather than claiming the diagnosis.
- **The GPU does not accelerate this workload.** Measured, in `analysis/device_sensitivity.json`: the GPU takes about 1.26x the CPU's wall clock for the same training run. The profile says why: the simulator is a sequential Python loop and is about 85% of a run, the network is about 8% of it, and on the GPU even that 8% is several times slower in absolute terms (tiny tensors, launch overhead). The measurement is in the artefact, and the earlier claim that the choice "costs about one percent" was false and has been removed from the code and from this file.

7 The artefacts

Every number above is read from one of these files, and the paper is generated from the same ones.

results_main.csv	the main table: every controller, five training seeds for the learned one
results_ablation.csv	the component ablation, twelve variants
results_ablation_stress.csv	the ablation on the combined-stress scenario
results_scenarios.csv	the nine scenarios and their variants
analysis/ab_observation.json	the observation arms
analysis/ab_energy_management.json	the energy arms
analysis/ab_credit.json	the credit-assignment arm
analysis/ab_budget.json	the training-budget sweep
analysis/ab_map_resolution.json	the map-resolution sweep
analysis/claim_audit.json	the claim audit, item by item
analysis/connectivity_accounting.json	the connectivity accounting
analysis/corner_transfer.json	the corner certificate's transfer
analysis/alt_deployment.json	the alternative corrected deployment

8 Reproducing this report

```
cd ml_impl
python analysis/make_report.py     # this report
python analysis/assemble_results.py # the paper
python analysis/final_checks.py    # the gate
python analysis/claim_audit.py     # the audit
python tests/run_tests.py -q        # the tests
```